L26: Initialization Bias and Warmup Detection

Simulation Without a Black Box

Outline

Two Regimes of Simulation Analysis

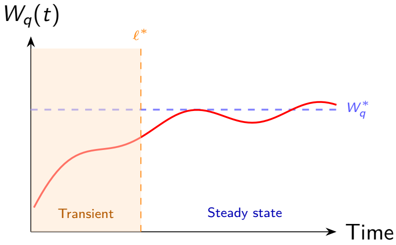
Terminating Simulation

The system has a natural stopping event (e.g., end of shift, clinic closes at 5 pm). Statistics are naturally bounded. Replications are the primary analysis tool.

Steady-State Simulation

We care about *long-run average* behavior ($t \to \infty$). No natural stopping event. Examples: 24/7 call center, continuous manufacturing line.

The warmup problem only arises in steady-state studies. Starting conditions (empty queue, idle servers) bias early observations — but the bias decays as the system approaches stationarity.



Initialization Bias: The Empty-and-Idle Problem

Why starting empty is almost always wrong:

- Real systems are rarely observed at time 0 from rest
- An M/M/1 queue at $\rho = 0.85$ has $L_q \approx 4.8$ in steady state, but starts at $L_q = 0$
- Including transient observations *downward-biases* $\bar{W}_q$

Magnitude of the bias:

ρ	True W_q^*	Bias without warmup
0.50	2.0 min	-0.3 min (-15%)
0.75	6.0 min	-1.8 min (-30%)
0.90	18.0 min	-9.2 min (-51%)

Warning

Bias grows rapidly as $\rho \to 1$. High-utilization systems require long warmup periods.

```python
import simpy, numpy as np

def mm1(lam, mu, seed, T_total):
    rng = np.random.default_rng(seed)
    env = simpy.Environment()
    server = simpy.Resource(env, 1)
    waits = []

    def customer():
        with server.request() as req:
            t0 = env.now
            yield req
            waits.append(env.now - t0)
            yield env.timeout(
                rng.exponential(1/mu))

    def arrivals():
        while True:
            yield env.timeout(
                rng.exponential(1/lam))
            env.process(customer())

    env.process(arrivals())
    env.run(until=T_total)
    return waits
```

Welch's Algorithm: Moving-Average Smoother

Welch's Procedure

1. Run R replications, recording observations $Y_{r,j}$ (r = rep, j = observation index)
2. Average across reps: $\bar{Y}_j = \frac{1}{R}\sum_r Y_{r,j}$
3. Apply moving average with window w:

$$\tilde{Y}_j = \frac{1}{2w+1}\sum_{i=-w}^{w} \bar{Y}_{j+i}$$

4. Find ℓ^* where $\tilde{Y}_j$ enters a $\pm 10\%$ band around the grand mean and stays there.

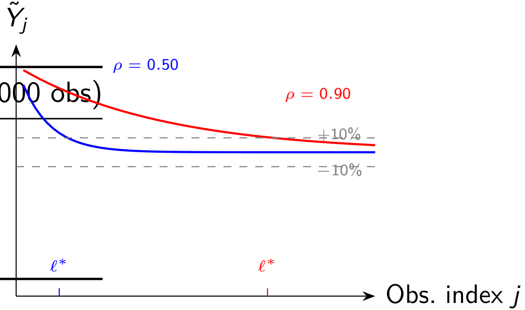
Recommended: $R \geq 5$ replications; $w \approx n/10$.

```python
import numpy as np

def welch_smoother(batch_means, w=50):
    """
    batch_means: 2D array (R, n)
    w: moving-average window
    """
    Y_bar = batch_means.mean(axis=0)
    n     = len(Y_bar)
    Y_tilde = np.convolve(
        Y_bar,
        np.ones(2*w+1)/(2*w+1),
        mode='valid'
    )
    return Y_tilde

# Find warmup: first index inside +-10%
def find_warmup(Y_tilde, tol=0.10):
    grand = Y_tilde.mean()
    inside = np.abs(Y_tilde-grand) \
             < tol*grand
    idx = np.argmax(inside)
    return idx
```

Effect of ρ on Warmup Length

Key observation: Higher utilization $\Rightarrow$ longer transient $\Rightarrow$ more data must be discarded.

ρ	Approx. warmup ℓ^*	% data discarded (5000 obs)
0.50	200 obs	4%
0.75	600 obs	12%
0.85	1 400 obs	28%
0.90	3 000 obs	60%



Practical consequence: Run the simulation *much longer* at high utilization. Warmup is not a fixed constant.

Warmup Deletion: Before vs. After

What changes when we delete the warmup?

Statistic	Without deletion	With deletion ($\ell^* = 1400$)
$\bar{W}_q$	12.1 min	18.4 min
95% CI half-width	1.2 min	1.4 min

The deleted estimate (18.4 min) is closer to the true $W_q^* = 18.0$ computed from the Pollaczek-Khinchine formula.

Variance trade-off: Deleting observations reduces n, slightly widening the CI. This is acceptable — bias is far more dangerous than a modest increase in variance.

```
# After collecting warmup array:
all_waits = mm1(lam=0.85, mu=1.0,
                seed=42, T_total=1e6)

ell_star = 1400    # from Welch plot

# Delete warmup observations
steady_waits = all_waits[ell_star:]

import numpy as np, scipy.stats as st

n    = len(steady_waits)
xbar = np.mean(steady_waits)
s    = np.std(steady_waits, ddof=1)
hw   = st.t.ppf(0.975, n-1)*s/n**0.5

print(f"Wq = {xbar:.2f} +/- {hw:.2f}")
```

Warning

Never use the moving-average window w as the warmup length. w is the smoother parameter; ℓ^* is the deletion point.

Key Takeaways

1. **Terminating vs. steady-state**: warmup analysis applies only to steady-state simulations.
2. **Initialization bias** from empty-and-idle can understate W_q by 15–51% depending on ρ.
3. **Welch's algorithm**: average across replications, apply moving-average smoother, find where $\tilde{Y}_j$ enters $\pm 10\%$ of the grand mean.
4. **Higher $\rho \Rightarrow$ longer warmup**: at $\rho = 0.90$, 60% of a 5000-observation run may need to be discarded.
5. **Bias beats variance**: deleting warmup slightly widens the CI but removes a large systematic error.
6. **Warmup is not fixed**: re-run Welch's procedure whenever the system design or ρ changes.

Next Lecture

L27 — Independent replications and confidence intervals.